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Exercise 1

SQL Fundamentals: SELECT & Filtering

01. Selecting & Sorting

01. SELECT *
FROM employees;

Table employees

id	first_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

02. SELECT DISTINCT department
FROM employees;

Department
IT
HR
Finance
Marketing

03. SELECT first-name,
last-name
FROM employees
ORDER BY salary DESC;

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. SELECT id,
first-name,
last-name,
salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02. Filtering with WHERE

05. SELECT id,
 first_name,
 last_name,
 department
 FROM employees
 WHERE department='IT';

id	first_name	last_name	department
5	Eva	Naidoo	IT
1	Alice	Green	IT

06. SELECT id,
 first_name,
 last_name,
 department,
 salary
 FROM employees
 WHERE department='Finance' AND salary > 60 000;

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

07. SELECT id,
 first_name,
 last_name,
 department
 FROM employees
 WHERE department='HR' OR 'Marketing'

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08. SELECT id,
 first_name,
 last_name,
 department
 FROM employees
 WHERE department = NOT 'IT';

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09. SELECT id,
 first_name,
 last_name,
 department
 FROM employees
 WHERE department = IN 'IT', 'HR', 'Finance';

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id,
 first_name,
 last_name,
 department,
 salary,
 city

FROM 'employees'
WHERE salary > 65000 AND city = 'Johannesburg'.

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	10000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg